

Real-Time Sign Language Interpretation

**Bridging Communication Gaps with
Innovative Technology**

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Introduction

Sign language interpretation is crucial for effective communication with the deaf and hard-of-hearing community. Despite its importance, there is a significant shortage of qualified interpreters, leading to communication barriers. Sign Language Interpreter, aims to address this issue by leveraging advanced technology to provide accessible and efficient interpretation services.

Motivation

5%

Use sign language globally

72%

Families don't sign with
Deaf children

75%

Deaf individuals face
job challenges

Need for real-time, AI-based sign language interpretation

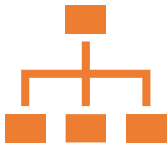


Goal

- Develop an AI-powered real-time sign language interpreter.
- Enable seamless communication without human translators.
- Improve accessibility for Deaf individuals.



Related work



Rule- based
systems



Glove based
systems



Computer vision
based approaches

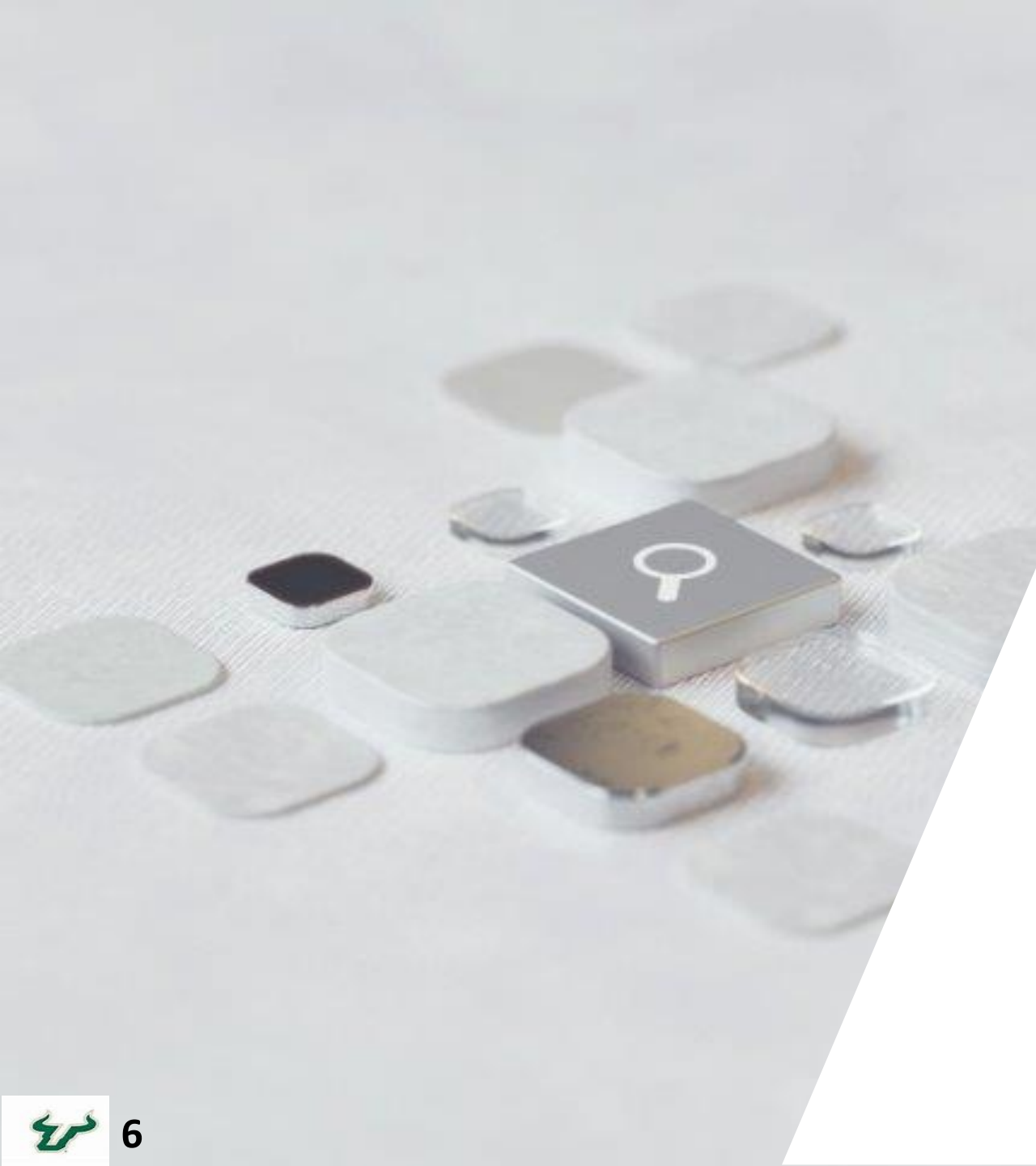


Existing Apps

Proposed solution

- Real-time Hand Gesture Recognition using CNN & OpenCV.
- Converts hand gestures into text & voice output.
- Offline Mode
- Scalable & Extendable
- Real-time processing for seamless communication.





Dataset

- ASL gesture dataset.
- Open-source repositories
- Custom recordings.
- LSA64



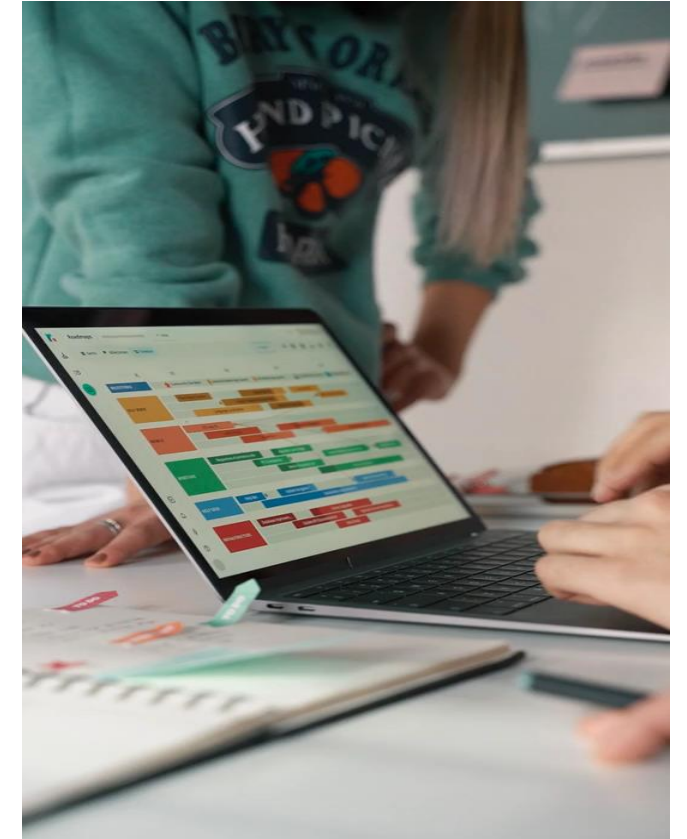
Data Preparation

- Data Collection
- Preprocessing
- Augmentation
- Splitting
- Feature Extraction

Model Selection

CNN

- Captures Spatial Relationships
- Automatic Feature Learning
- Scalability



- Python
- TensorFlow/Keras
- OpenCV
- SQLite
- pytsx3

Technologies and tools



Implementation



Setting Up Hand Histogram

Captures the skin tone histogram to segment hand gestures from the background.

Creating and labeling Gestures

Captures images of different hand signs and stores them in directory after labeling.



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Captures images of different hand signs and stores them in directory after labeling.

Data Augmentation

Enhances dataset by flipping images horizontally

Preprocessing and dataset preparation

Reads images, converts them to grayscale, and normalizes pixel values. Finally, splits the dataset.



Preprocessing and dataset preparation

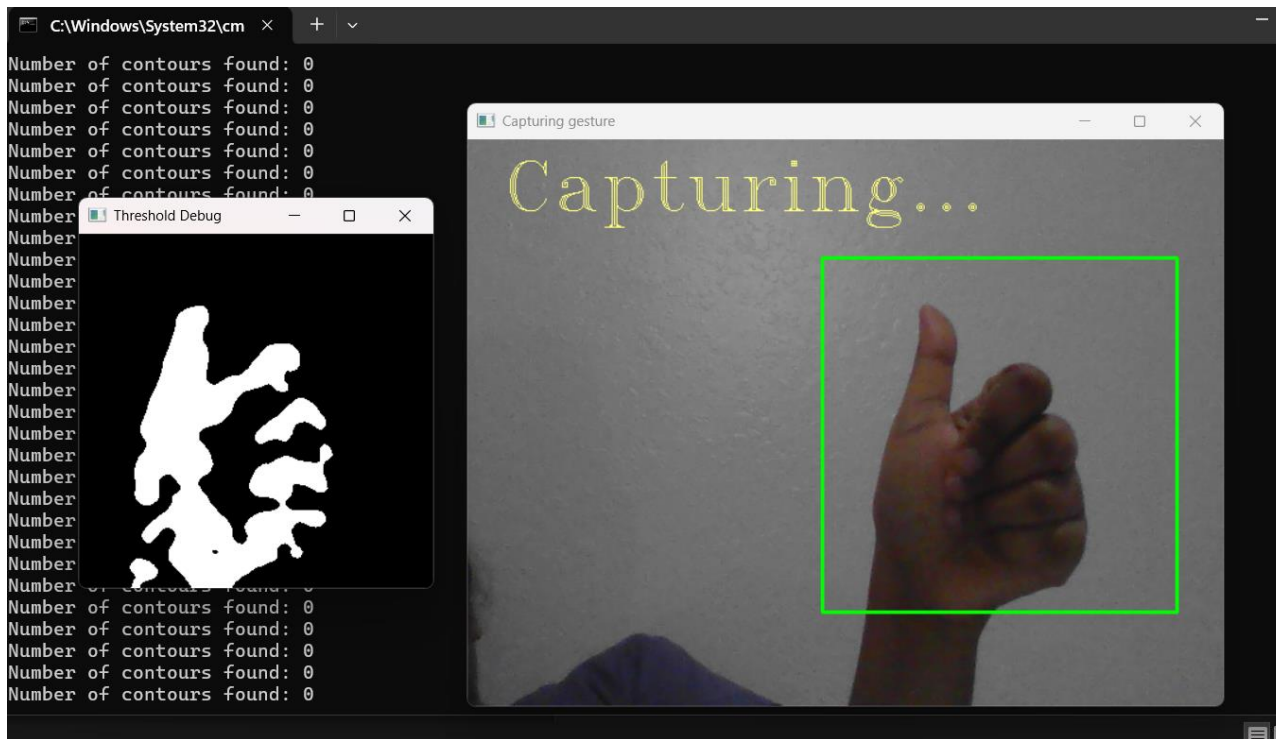
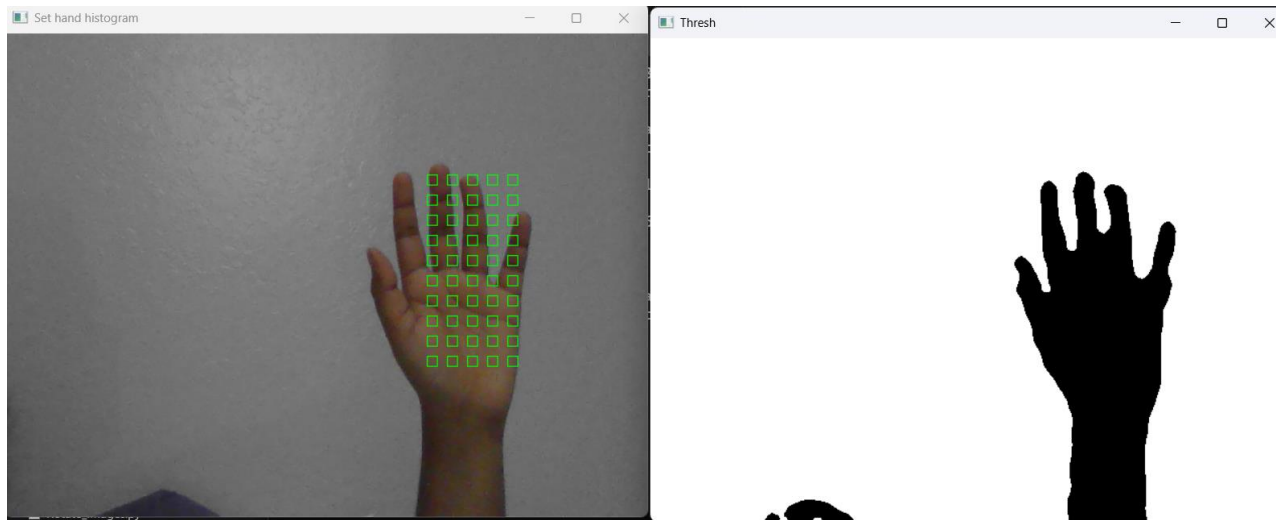
Reads images, converts them to grayscale, and normalizes pixel values. Finally, splits the dataset.

Training cnn model

Trains a CNN for sign language recognition and saves the best model in .h5 file.

Real-Time Gesture Recognition

Converts recognized gestures into text and speech output





Challenges

- Low accuracy on complex gestures
- Background Noise in Camera Feed
- Sign Variation Across Users

Solutions

- Increased dataset size & used data augmentation.
- Applied image thresholding & hand segmentation.
- Added multiple variations for each gesture.





Future Scope

- Mobile App Deployment for accessibility.
- Integration with Virtual Assistants (Alexa, Google Assistant).

Applications



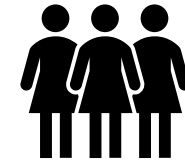
Healthcare

Assisting patients with disabilities.



Education

Helping students with hearing impairments.



Customer Service

Improving accessibility in businesses.



Conclusion

Achieves real-time gesture recognition with high accuracy using CNNs. It overcomes challenges through optimized training and data augmentation, making it scalable and accessible. Future enhancements will expand its global impact for inclusivity.

Thank You

